Johannes Roth

Machine Learning Engineer

johannes@roth24.de LinkedIn GitHub Scholar jroth.space Leipzig, Germany

Available from January 2027

Machine learning engineer with experience taking ML and data projects from initial requirements to deployment and operation. Background in production computer vision, large-scale data processing, and independent research in computational neuroscience.

EXPERIENCE

PhD Researcher - Computational Cognitive Neuroscience May 2022 – Present
Max Planck Institute for Human Cognitive and Brain Sciences & University of Giessen

- Conducted independent research at the intersection of **machine learning and computational neuroscience**, focusing on vision models and their relationship to human brain activity.
- Developed methods for model evaluation and experimental design, combining computational analyses with fMRI studies.
- Developed **GPU-based ML pipelines** for dataset curation and feature extraction, and released large-scale image datasets for computer-vision research.
- Independently developed and maintained research software and cloud-based data services for **LAION-fMRI** and **re:vision**, including public websites and annotation tools.
- Supervised working students and interns and collaborated with researchers across computer science and neuroscience.

Research Scientist - Machine Learning in Medicine Jun 2021 – May 2022
ScaDS.AI Dresden/Leipzig

- Developed and evaluated deep-learning models for **medical image analysis**, including brain-tumor segmentation.
- Built predictive models for clinical data, with a focus on mortality prediction and uncertainty estimation.

Data Scientist / ML Engineer (Working Student) Nov 2019 – May 2021
CHECK24, Travel Vertical

- Independently designed and deployed a **production computer-vision service** for **20 million hotel images**, covering model development and backend engineering.
- Prototyped additional computer-vision applications for image retrieval, duplicate detection, and image-quality assessment.
- Optimized recommendation-data pipelines through incremental processing, reducing update time by over **95%**.

Full-Stack Developer (Freelance) Oct 2020 – May 2021
Kimetric UG

- Designed and delivered Django-based websites for research organizations, including **Hebart Lab** and the **THINGS Initiative**.
- Gathered client requirements and managed deployment, Linux hosting, and ongoing application maintenance.

Data Scientist (Working Student) Oct 2018 – Sep 2019
Webdata Solutions GmbH (now Vistex)

- Developed machine-learning models for **product matching and image classification** in e-commerce.
- Improved image-based matching accuracy from **12% to 83%** against a classification baseline in an internal cross-retailer evaluation.

Data Analyst (Working Student) Feb 2018 – Aug 2018
Mercateo Services GmbH & Co. KG

- Developed data-integration, reporting, and testing workflows for an **AWS-based data warehouse** using Apache NiFi and Pentaho.

TECHNICAL SKILLS

- **ML & Analysis:** Python, SQL, PyTorch, TensorFlow, scikit-learn; computer vision, embeddings, model evaluation, statistical analysis, experimental design.
- **Software & Data:** FastAPI, Flask, Django, Redis, Docker, Git, CI/CD; data pipelines, similarity search, dataset curation.
- **Cloud & Compute:** AWS (S3, IAM), Linux, SLURM/HPC, multi-GPU processing.
- **Languages:** German (native), English (fluent).

OPEN SOURCE

- Contributed to **thingsvision**, an open-source library for computer-vision research, with work on software architecture, model integration, and memory efficiency.

EDUCATION

Doctoral Candidate (Dr. rer. nat.) May 2022 – Present
University of Giessen
Research focus: machine learning and computational neuroscience.

M.Sc. Computer Science 2017 – 2021
Leipzig University; grade 1.2

- **Master's thesis (grade 1.1):** Applied generative deep learning to study visual processing in the human brain.

B.Sc. Business Information Systems 2014 – 2017
Leipzig University; grade 1.5

SELECTED PUBLICATIONS

- F. P. Mahner*, **J. Roth***, et al. *Characterizing Universal Object Representations Across Vision Models*. **NeurIPS 2026** (accepted). Shared first author.
- **J. Roth***, Y. Duan*, et al. *Ten principles for reliable, efficient, and adaptable coding in psychology and cognitive neuroscience*. Communications Psychology, 2025. Shared first author.
- **J. Roth** et al. *Multi-plane UNet++ Ensemble for Glioblastoma Segmentation*. MICCAI BrainLes 2021 workshop proceedings, published 2022.

RECOGNITION & SERVICE

- Presented research at **CCN 2025 (oral)**.
- Elected representative for **180+ doctoral researchers** at MPI CBS (2023–2024).
- **CMBB Replication Award (2025)** for contributions to reliable coding practices in neuroscience.